

INTELLIGENCE GATHERING REPORT

Passive Reconnaissance & OSINT Analysis

Prepared By

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Course: CEH (Certified Ethical Hacker)

Institute: BETS

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1. Executive Summary

This report documents the results of a passive information gathering exercise conducted against the domain google.com as part of the CEH (Certified Ethical Hacker) coursework at BETS. The objective was to enumerate publicly available intelligence including domain ownership, IP address resolution, open ports, exposed services, and potential attack surface indicators — without sending any intrusive traffic to the target.

All activities were performed using legal, passive OSINT tools including WHOIS, nslookup, Shodan, and Google Dork queries. No active exploitation was attempted.

2. Target Overview

Target Domain	google.com
Investigation Date	April 16, 2026
Investigator	Ahmad Shoukat
Tools Used	WHOIS, nslookup, Shodan, Google Dorks
Scope	Passive reconnaissance only — no active scanning
Classification	Academic / Ethical Hacking Lab Exercise

3. WHOIS / Domain Ownership Analysis

The WHOIS query was executed using the Kali Linux terminal command `whois google.com`. The results returned two records — one from VeriSign's registry database and one from MarkMonitor's registrar database.

3.1 Domain Registration Details

Domain Name	GOOGLE.COM
Registry Domain ID	2138514_DOMAIN_COM-VRSN
Registrant Organization	Google LLC
Registrant Country	United States (US)
Registrar	MarkMonitor, Inc.
Registrar IANA ID	292
Registrar URL	http://www.markmonitor.com
WHOIS Server	whois.markmonitor.com
Creation Date	September 15, 1997
Last Updated	August 2, 2024
Registry Expiry Date	September 14, 2028
DNSSEC Status	Unsigned

3.2 Domain Status Flags

The domain has all six standard protection flags applied, indicating a high-security registration profile:

- `clientDeleteProhibited` — Prevents deletion by the registrar
- `clientTransferProhibited` — Prevents transfer to another registrar
- `clientUpdateProhibited` — Prevents unauthorized updates
- `serverDeleteProhibited` — Registry-level delete lock
- `serverTransferProhibited` — Registry-level transfer lock
- `serverUpdateProhibited` — Registry-level update lock

The presence of all six flags confirms that `google.com` is fully locked against hijacking, unauthorized transfer, or modification — typical of a high-value enterprise domain.

3.3 Authoritative Name Servers

- `ns1.google.com`
- `ns2.google.com`
- `ns3.google.com`
- `ns4.google.com`

Google operates its own authoritative DNS infrastructure. All four name servers are self-hosted under the `google.com` namespace, which is standard practice for large enterprises with global DNS requirements.

4. DNS Resolution & IP Address Analysis

The `nslookup` command was used to resolve DNS records for `google.com`. The query was directed to the local DNS resolver at `192.168.169.2` (a non-authoritative resolver, likely the default gateway or ISP DNS server).

4.1 Resolved Addresses

Query Server (Resolver)	192.168.169.2
Answer Type	Non-authoritative (cached by local resolver)
IPv4 Address (A Record)	142.250.200.174
IPv6 Address (AAAA Record)	2a00:1450:4019:815::200e

4.2 Analysis Notes

- The IPv4 address 142.250.200.174 belongs to Google's global anycast infrastructure. The IP resolves differently by geographic region due to Google's use of load balancing and CDN routing.
- The IPv6 address 2a00:1450:4019:815::200e is from Google's 2a00:1450::/32 prefix, which is assigned to Google's European infrastructure. This suggests the DNS resolution was routed through a European PoP at the time of the query.
- The response was non-authoritative, meaning the resolver returned a cached answer from Google's own name servers rather than querying them directly.

5. Shodan Analysis — IP 73.50.186.152

A Shodan lookup was performed to analyze the host at IP address 73.50.186.152. This IP was investigated as part of the reconnaissance lab exercise to practice interpreting Shodan data for open port and service enumeration.

5.1 Host General Information

IP Address	73.50.186.152
Hostname (PTR)	c-73-50-186-152.hsd1.il.comcast.net
Domain	comcast.net
Country	United States
City	Chicago, Illinois
Organization	Comcast IP Services, L.L.C.
ISP	Comcast Cable Communications, LLC
ASN	AS7922
Operating System	Synology DiskStation Manager (DSM) 7.1.1-42962
Last Seen on Shodan	April 16, 2026

5.2 Open Ports & Services

Shodan identified the following open ports on this host. The presence of multiple management and application ports on a residential Comcast IP is notable.

Port / Protocol	Service	Version / Banner	Risk Level
22 / TCP	SSH	OpenSSH 8.2 (ssh-rsa)	Medium
80 / TCP	HTTP	Web server (unencrypted)	Medium
443 / TCP	HTTPS	TLS/SSL encrypted web	Low

5000 / TCP	DSM Web UI	Synology DiskStation Manager	High
5001 / TCP	DSM HTTPS UI	Synology DSM (HTTPS)	Medium
8000 / TCP	HTTP Alt	Alternate HTTP service	Medium
8001 / TCP	HTTP Alt	Alternate HTTP service	Medium

5.3 SSH Key Fingerprint

Shodan captured the SSH host key during its last scan. The key details are:

- Key Type: ssh-rsa
- Fingerprint: e3:40:3b:6a:64:fb:b7:4d:18:26:61:81:ae:83:32:07
- KEX Algorithms: curve25519-sha256, curve25519-sha256@libssh.org

The SSH fingerprint can be used for host verification. Its public exposure in Shodan means it is known to threat actors — increasing the risk of targeted brute-force or key-related attacks.

6. Google Dorking Results

Google Dork queries were used to extract publicly indexed information about targets. Two dorks were executed during this exercise.

6.1 site: Operator — site:www.google.com

This dork enumerates publicly indexed pages under the www.google.com domain. Results included:

- Google Doodles page (<https://www.google.com/doodles>)
- About Google page (<https://www.google.com/about>)
- Google NotebookLM AI tool (<http://www.google.com/notebook>)
- Google Chrome download page (<https://www.google.com/chrome>)
- Brand Resource Center (<https://www.google.com/permissions>)

From a reconnaissance perspective, this dork reveals the public-facing web structure and subpaths of the target. In a penetration test, these paths can be used to map the application surface.

6.2 intitle: Operator — intitle:index.of metasploitable

This dork searches for open directory listings containing the keyword 'metasploitable'. It is used in ethical hacking labs to locate exposed file servers that host vulnerable VM images. Results included:

- Southern University of Science and Technology (sustech.edu.cn) — exposed /cs315/2025Fall/ directory containing metasploitable-linux VM image files
- Saint Mary's University of Minnesota (cs.smumn.edu) — exposed /courses/ISO/ directory with metasploitable-linux-2.0.0.zip and Metasploitable 3 workspace
- Universidad Politecnica de Madrid (vnx.dit.upm.es) — /vnx/filesystems/ directory with various VM images

These findings demonstrate how misconfigured web servers expose sensitive lab materials. The metasploitable VM images themselves are intentionally vulnerable

systems — their public exposure is low risk in isolation but indicates poor web server security hygiene at the hosting institutions.

7. Suspicious Indicators & Risk Assessment

7.1 google.com Domain

Indicator	Severity	Notes
DNSSEC is unsigned	Low	Vulnerable to DNS spoofing in theory; Google mitigates this via infrastructure controls
Registrant email hidden behind MarkMonitor privacy form	Informational	Standard GDPR/ICANN privacy proxy; not a vulnerability

7.2 Shodan Host 73.50.186.152 — Key Risks

- NAS Device Exposed to Internet: A Synology DiskStation Manager (DSM) running on a residential Comcast IP is a high-risk configuration. DSM has a history of CVEs, and exposing it directly to the internet greatly increases the attack surface.
- Port 5000 Open (DSM Admin UI): The Synology DSM admin interface is exposed on port 5000 without TLS. Accessing NAS admin panels over cleartext HTTP is a critical security misconfiguration.
- SSH on Port 22 Publicly Exposed: OpenSSH 8.2 is accessible publicly. While OpenSSH 8.2 itself is reasonably patched, exposing SSH on a residential connection without IP whitelisting invites brute-force attempts.

- SSH Key Fingerprint in Shodan: The RSA host key fingerprint is indexed publicly, meaning any attacker can verify they are talking to this exact host during a MITM attempt.
- Multiple Alternate HTTP Ports: Ports 8000 and 8001 suggest additional web applications running on this host. These services may host internal tools, dashboards, or APIs not intended for public access.

8. Conclusions & Recommendations

This exercise demonstrated the effectiveness of passive reconnaissance techniques in building a comprehensive intelligence profile of a target without direct interaction. The key takeaways are:

8.1 Findings Summary

- google.com is owned by Google LLC, registered through MarkMonitor with full domain lock protections — no attack surface found at the domain layer.
- DNS resolves to Google's global anycast IP infrastructure with both IPv4 and IPv6 records. The IPv6 address suggests European CDN routing.
- The Shodan-analyzed host (73.50.186.152) presents a significantly more concerning exposure profile — a Synology NAS with admin UIs, SSH, and multiple HTTP services exposed to the public internet on a residential ISP connection.
- Google Dorks revealed that university web servers are misconfiguring directory listings and exposing Metasploitable VM images — an indicator of inadequate web server hardening.

8.2 Recommendations

- For the Shodan host: Immediately firewall port 5000 from public access, enforce HTTPS-only on DSM (port 5001), restrict SSH access to known IPs using firewall rules, and review all services running on ports 8000/8001.
- For the university servers: Disable directory listing (Options -Indexes in Apache, autoindex off in Nginx), move sensitive files behind authentication, and conduct periodic web server audits.
- General: Enable DNSSEC on domains where applicable. Use a secrets management approach rather than exposing admin interfaces to the public internet.

9. Tools & Methodology

Tool	Type	Purpose
whois	CLI / Passive	Domain registration, ownership, and registrar data
nslookup	CLI / Passive	DNS A and AAAA record resolution
Shodan (shodan.io)	Web OSINT / Passive	Open port enumeration, banner grabbing, service fingerprinting
Google Dorks	Web OSINT / Passive	Indexed page enumeration, open directories, information leakage
Kali Linux 2024	Platform	Operating environment for all CLI tools

— END OF REPORT —

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